

Introduction Slides

14 January 2021 05:10

1. IP protocol model (similar to OSI protocol layers)

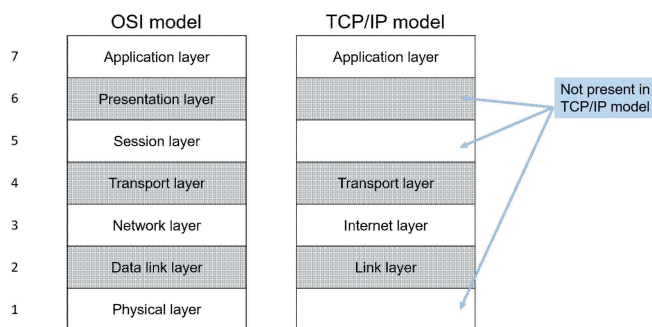
a. What is the purpose of layering or protocols?

Ans:

- Each component handles a different part of the communication.
- to break up the sending of messages into separate components and activities.
- each layer deals with one functional aspect of the communication

b. Draw the layered architecture (IP or OSI)!

Ans:



c. What's the difference between a protocol function and a service of a protocol?

Ans:

Services send and receive data via protocols. A service manipulates data, a protocol moves it around. A Service provides a functionality to the above layers. A protocol is a set of rules to be followed within a layer.

2. What's the functionality provided by the (a) network layer, (b) transport layer, (c) link layer and (d) physical layer?

Ans: **Network Layer**

Features:

1. Transports packets across network
2. Fragmentation
3. Adds the Source and destination address
4. Routing

Services: 1. Packetization, Routing and Forwarding.

Transport Layer

Features:

1. Service point addressing - Port
2. Error Control - through re-transmission
3. Flow control
4. Connection control - connection and connectionless oriented.
5. Segmentation and reassembly - replaces packets which were lost

Link Layer

Features:

1. Framing - packets from network layer to frames and then sends bit by bit on physical layer. At receiver, takes each bit and makes frames.
2. Addressing - hardware
3. Synchronization of data frames.
4. Error Control - Cannot ask for retransmission, drops corrupted frames.
5. Multi-Access - provides CSMA/CD - to access a shared medium.

Physical layer

Features:

1. Converts frames from link layer to electric signals to be sent over physical medium

2. Data to signal coding,

3.What's the difference between the transport protocol and the logical link sublayer?

Ans: The Logical Link Control (LLC) sublayer provides the logic for the data link, thus it controls the synchronization, flow control, and error-checking functions of the data link layer.

4.What is the purpose of a TCP/UDP port and what is the difference to the IP address?

Ans: TCP/UDP port is used for Application to Application communication.

IP address is used to identify the destination machine on which the target application resides.

5.What does CSMA stand for? What's the idea and how does it work?

Ans: CSMA is Channel Sense Multiple Access. The idea is to sense the channel to see if its idle before transmitting the data. If the Channel is busy, transmission is deferred for a random amount of time. By doing this the collision from data transmission from multiple channels using the same medium is minimized.

6.What's the purpose of IP subnetting? What's the effect on routing?

Ans: Purpose is to reduce the usage of IP addresses in the network with the same number of devices such that the traffic on the main network is less congested making the network more efficient.

Through subnetting, network traffic can travel a shorter distance without passing through unnecessary routers to reach its destination.

When a router interface is configured with an IP address and subnet mask, the interface becomes a host on that attached network. A directly connected network is a network that is directly attached to one of the router interfaces. The network address and subnet mask of the interface, along with the interface type and number, are entered into the routing table as a directly connected network.

7.What's the difference between MAC and IP addresses?

Ans:

Sr. No.	Key	MAC Address	IP Address
1	Definition	MAC Address stands for Media Access Control Address.	IP Address stands for Internet Protocol Address.
2	Usage	MAC Address ensure that physical address of the computer is unique.	IP Address is a logical address of the computer and is used to uniquely locate computer connected via a network.
3	Format	MAC Address is of six byte hexadecimal address.	IP Address is of 4 bytes or of 16 bytes.
4	Access Protocol	MAC Address can be retrieved using ARP protocol.	IP Address can be retrieved using RARP protocol.
5	Provider	Chip maker manufacturer provides the MAC Address.	Internet Service Provider, ISP provides the IP Address.

8.What does VPN mean? Which protocol implements it?

A VPN (virtual private network) is a service that creates a safe, encrypted online connection. Internet users may use a VPN to give themselves more privacy and anonymity online or circumvent geographic-based blocking and censorship. Explanation: There are six types of protocols used in VPN. These are Internet Protocol Security or IPSec, Layer 2 Tunnelling Protocol (L2TP), Point – to – Point Tunnelling Protocol (PPTP), Secure Sockets Layer (SSL), OpenVPN and Secure Shell (SSH).

9.What is the difference between VoIP and circuit-switched voice?

VoIP uses broadband connection to transmit data but circuit switched voice networks use an old technology of copper wires connected between two ends forming a complete circuit before communication starts.

Wireless Transmission

16 February 2021 11:30

RADIO FREQUENCIES AND SIGNALS

1. What are the wavelengths of the different band used by GSM and WLAN?

GSM Frequencies - 900MHz and 1800MHz (33.3 cm and 16.7 cm)

WLAN frequencies - 2.4GHz and 5 GHz (12 cm and 6.0 cm)

2. Which bands typically provide better penetration of walls?

Bands with low frequencies and higher wavelengths provide better penetration through walls.

Ex: VHF/UHF-ranges for mobile radio (< 3Hz)

3. What's the disadvantage of using higher frequencies for communications? What's the advantage?

Disadvantage - No penetration of walls, fading, pathloss, etc.

Advantages - higher bandwidth, small antennas.

4. Sketch the frequency spectrum of a 100 MHz sinus signal and of a symmetric rectangular signal with the same period.

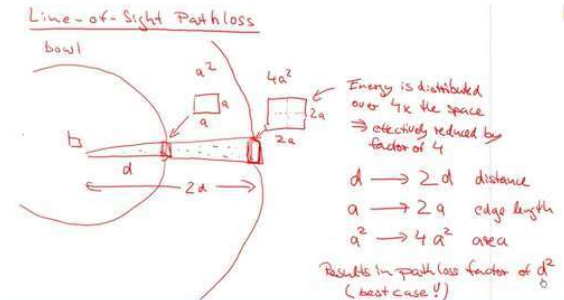
Frequency spectrum - discrete

Rectangular wave - find time period of this from $c = f \cdot \lambda$ (of 100MHz)

$T = 1/F$

SIGNAL PROPAGATION

1. Why is a pathloss proportional to d^2 is assumed for line-of-sight connections? Why is d^2 considered as the best case?



It is considered as the best case because the calculations are made not taking into account any other interferences. It's the strength of the signal that is resulting only from the distance and no other factor.

2. What are the reasons for fast fading? What is exactly happening?

Fast fading generally occurs with mobility, because of which the channel characteristics change with time and location - signal paths change, different delay variations of different signal parts, different phases of signal parts resulting in fast fading or short-term fading.

3. Why does fast fading depend on mobility? Why is fast fading frequency dependent?

Due to mobility, channel characteristics change with time and location.

When a user (or reflectors in its environment) is moving, the user's velocity causes a shift in the frequency of the signal transmitted along each signal path. Since fading depends on whether signal components add constructively or destructively, frequency plays an important role.

4. What is the difference between fading due to simple phase shifts between arriving signals and inter symbol interference?

ISI - fading caused due to time shift reception.

Simple phase shift - fading due to phase shift (Guard space is taken over by the symbol)

5. What are the factors of the received signal important for error-free reception?

- the signal to interference & noise ratio SINR is above a threshold (this depends on the radio technology (modulation, coding), HW and signal processing capabilities of the receiver)
- The received signal strength S is above a threshold (this is defined by the sensitivity of the receiver)

Antennas

1. What are the characteristics of an isotropic radiator?

- Have equal radiations in all directions (3-dimensions)
- This is only a theoretical concept
- Real antennas have directional effect.

2. What directional effects show simple practical antennas?

In a simple practical antenna or a dipole, the antenna gain is maximum towards the main lobe compared to the power of an isotropic radiator where the gain is equal in all directions.

3. How can you build a directional antenna from 2 simple $\lambda/2$ antennas? What are the prerequisites?

Modulation and Coding

1. How can we add information to a periodic signal? What are the choices?

By modulating the periodic signal, information can be added to it. The choices are:

- Amplitude Modulation (AM)
- Frequency Modulation (FM)
- Phase Modulation (PM)

2. Why is amplitude modulation in its pure form rarely used in practice?

When does it make sense? Where is the idea applied and what is the benefit of its application?

Verz susceptible to interference.

3. What is the advantage of 16QAM over QPSK?

- Less errors compared to PSK.
- More band width as single symbol can carry more bits.

4. What limits the use of higher order modulation schemes, as e.g. 16QAM or 64QAM?

Error Rate increases with higher order modulation schemes.

5. How can we optimize the throughput of a single transmitter-receiver pair? How can we optimize it under given bandwidth constraints as well as fixed signal and interference values?

By having a optimal trade off between modulation and coding schemes of the signals and also the power of the signal.

Multiplexing – Splitting up the Radio Resource

1. What is the purpose of multiplexing?

Multiple use of a shared medium to provide it to multiple entities rather than just a single one

2. What kind of multiplexing is used for public FM radio?

Frequency division multiplexing.

3. Which kind of multiplexing is used for GSM?

Time and frequency multiplexing

4. What is the problem of multiplexing in time, what is its advantage?

Precise synchronization needed

Advantages:

- only one carrier in the medium at any time
- throughput high even for many users

Spectrum Spreading

1. What is the benefit of spectrum spreading? What is needed for spreading?

- It helps protect the narrow band signal from fading due to narrow band interference (frequency selective fading)
- Spreading the narrow band signal into a broadband signal using a special code

2. What is the benefit of frequency hopping? What are its problems?

- Frequency selective fading and interference limited to short period
- Simple implementation
- Uses only small portion of the spectrum at a time.
- Problem - not robust as DSSS, simpler to detect.

Cellular Networks – Capacity and Interference

1. Spatial reuse in cellular systems: compare a reuse of 3 with a reuse of 9 with respect to capacity (data rate) and quality (error rate).

3 reuse factor - high data rate, low quality (higher error rate due to low SIR)

9 reuse factor - low data rate, high quality (lower error rate due to high SIR)

2. What is the effect of increasing the transmit power of all base stations in interference-limited systems?

Increasing the power of a BS causes interference to neighbouring cells resulting them to increase in power which again causes interference. Hence increasing power after a optimum point induces lot of noise.

3. How can we increase the system capacity in interference-limited cellular systems? And how not? Discuss the effects of different approaches.

- By an optimal combination of modulation and coding schemes.
- By not increasing the power only.
- By increasing the modulation order, higher data rate will be supported but increasing it above a optimum induces higher error rate. Increase of power already discussed. And increase of coding schemes might reduce error rate but there is an overhead of bandwidth for supporting the redundancy.

4. What are the results of adding more base stations to a given cellular network layout assuming that there is no extra spectrum available? What are the measures that need to be taken to handle this?

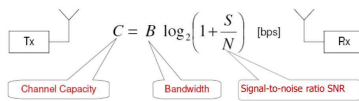
- What are the effects of increasing the spatial separation, e.g. employing 6 instead of 3 sectors per base station?
- Discuss how direct communication between mobiles (D2D) is a solution to

improve the network capacity of cellular networks? What are the gains of this?

Channel Capacity – Signal vs. Interference

1. How do interference and the received signal strength determine the capacity of a channel?

Using Shannon's channel capacity equation.



2. Why is interference – or more accurately its absence – as important as the signal itself?

Even when the signal strength is high, if there is too much interference caused due to limited resources and high users, same channels will be used by many users causing higher error rates.

3. What defines an interference-limited system? What happens there? How can we increase the system capacity in these systems? And how not?

- Mobile station is within coverage zone
- Signal is sufficient, but too much interference I at the receiver, SINR is too low.
- SINR is too low
- No more resources / capacity left
- By Increasing the Bandwidth. Eg: by employing spread spectrum technologies.
- By NOT using lower order modulation schemes or By using low bandwidth.

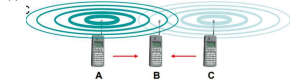
Medium Access Schemes

22 February 2021 15:37

Basics of Wireless Access Control

1. **What is the responsibility of the Media Access Control sublayer in communication systems?**
MAC layer is responsible for sending and receiving data over a shared channel by some strategies like Multiple Access schemes
Carrier sensing
Exchange of control packets
Collision detection
MAC has error detection capabilities through cyclic redundancy checks and Parity schemes.
Flow control
Power management
Mobility issues.
Used To control Medium access dynamically
2. **What are the new problems wrt MAC in wireless systems?**
Hidden terminal problem
Near and far terminal problem
Exposed terminal problem
3. **What are the causes of the problems in wireless systems?**
 - The signal strength decreases (at least) proportional to the square of the distance resulting in inconsistent views between sender and receiver
 - The sender would apply CS and CD, but the collisions happen at the receiver
 - It might be the case that a sender cannot "hear" the collision, i.e., CD does not work
1. **Why does CSMA/CD known from Ethernet fail?**
 - The signal strength decreases (at least) proportional to the square of the distance resulting in inconsistent views between sender and receiver
 - The sender would apply CS and CD, but the collisions happen at the receiver
 - It might be the case that a sender cannot "hear" the collision, i.e., CD does not work

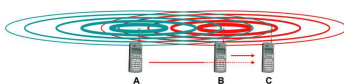
2. **What is the hidden terminal problem? When does it happen? What happens?**



3. **What is the exposed terminal problem? When does it happen? What happens?**



4. **What is the near far problem? What happens there?**



5. **What are the dimensions in which we can separate the access?**
Space, time, frequency, code

Code Division Multiple Access

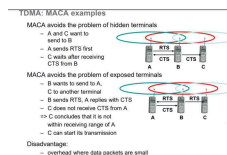
1. **What is the trick with CDMA to avoid collisions?**
Each sender XORs the data to be sent with a unique code on the same shared channel.
The receiver which needs this data will TUNE into this using the unique code of sender.
2. **Why is power control important for correct functioning of CDMA?**
Because all terminals send data via the same channel with the same frequency, if the power of Particular transmission is less, it fades away due to near far problem.
all terminals can use the same frequency, less planning needed
all signals should have the same strength at a receiver
3. **What happens if the keys are not orthogonal?**
It would be difficult to despread the signals at the receivers with the non orthogonal codes.
Interference would be noticed between data signals.
4. **What is the advantage of CDMA with respect to the handling of noise?**
The white noise (Or interference) is not coded.
5. **Does CDMA require the same, higher or lower SINR for correct transmission? Why?**

Multiplexing and Duplexing

1. **What is simplex communication? What are example systems for this?**
– unidirectional link transmission
Exmples - radio broadcasting, television broadcastin
2. **What are the pros and cons of TDD over FDD?**
+ support for asymmetric traffic
- mix of uplink and downlink interference on single band
Examples: DECT, WLAN
3. **For which of the given cases is FDD preferable? Why?**
4. **What does uplink and downlink mean?**
Uplink and downlink, also called upload and download, refers to the two-way communication between a cell tower and your phone. Downlink definition – signal coming from a cell tower to your cellular device. Uplink definition – signal leaving your cellular device and going back to a cell tower,

Time Division Multiple Access

1. **What is the idea of ALOHA?**
Idea is to randomly access the medium when a station wants to send data on the shared channel.
 2. **Why does slotted ALOHA outperform ALOHA with respect to throughput?**
The Slotted ALOHA is better than the Pure ALOHA. As the probability of collision is less in Slotted ALOHA as compared to Pure ALOHA because the station waits for the next time slot to begin which let the frame in a previous time slot to pass and avoids the collision between the frames.
 3. **What does implicit reservation mean?**
 - a certain number of slots form a frame, frames are repeated
 - stations compete for empty slots according to the slotted aloha principle
 - once a station reserves a slot successfully, this slot is automatically
 - assigned to this station in all following frames as long as the station has data to send (implicit reservation)
 - competition for these slots starts again as soon as the slot was empty in the last frame
 4. **How does explicit reservation work?**
Two modes:
 - ALOHA mode for reservation: competition for small reservation slots, collisions possible
 - reserved mode for data transmission within successful reserved slots (no collisions)
- Synchronisation:** it is important for all stations to keep the reservation list consistent at any point in time and, therefore, all stations have to synchronize from time to time
5. **What is the difference between random access and reservationbased access? What are the pros and cons of the two strategies?**
In random access a node can access the channel when it has data to send. Here collision chances are more. Fast and effective in case of low resource usage.
In reservation based access, a future slot should be reserved by the station and can send the data in that slot only. Collision is minimum. Overhead due to reservation process.
 6. **Compare the access latencies of both access schemes. Consider the latency under low and high loads.**
Same as above
 7. **Compare the two access strategies with respect to the maximum load they can handle.**
Same as above
 8. **Name the limits of distributed approaches to media access?**
 9. **How can the hidden-terminal problem be solved? Does the RTS/CTS scheme (MACA) solve the problem? In any case?**



Basic difference between Explicit reservation and reservation TDMA

WLAN 802.11

26 February 2021 17:52

Basics

• What is the difference between infrastructure and ad-hoc mode?

In infrastructure mode, stations communicate with each other via an access point that is connected to a portal via a distribution system which in turn is connected to other wired networks.'

In ad-hoc communication, there is no access point and stations directly communicate with each other.

• What is the purpose of the distribution system?

It forms an interconnection between all the access points in order to form a logical network based on several BSS (Basic Service Sets).

• Which protocol layers does the 802.11 standard comprise?

Physical layer and the Data Link layer

• What are the main differences between the different PHY variants?

Modulation, Bandwidth or data rates supported, frequencies used.

MAC

1. • Exponential backoff: what does it mean? What is its purpose?

It is a method implemented for fairness in access to shared medium where every station waits for a random amount of time (0: CW) after the IFS to send the data.

It is a collision avoidance mechanism.

2. • Describe DCF and PCF. What are the differences, pros and cons of each approach?

DCF (Distributed Co-ordination Function) is used for Asynchronous medium access while PCF is used for Polling all stations (by Point Co-ordinator).

1. • DCF: what is the purpose of different interframe spaces?

Signifies different packet priorities, smaller the IFS, higher the priority of that packet.

2. • Assume a node in DCF mode with RTS/CTS activated wants to transmit multiple segments of data: are multiple RTS/CTS cycles required for this?

No, only one RTS-CTS cycle for all the data fragments.

3. • For which messages does the SIFS apply? For what reason?

For ACK and CTS responses. Because these have highest priority.

4. • What is the purpose of the NAV?

It is used to determine if the medium is busy or not. If the NAV counter is zero, then the medium is free, else it's busy.

1. • How does DCF handle situations with high load? What are the strategies?

When there is high load, there is a higher chance of collision even with the random back-off timer, hence to handle this, the contention window size is doubled on every collision. Retransmission.

MAC Management

1. How does beaconing work in ad-hoc mode?

Beacon may be sent by any station (sending of beacon employs a random delay to avoid collisions)

2. • How are collisions among beacons avoided in ad-hoc mode?

Sending of beacon employs a random delay to avoid collisions

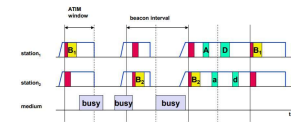
3. • What's the idea behind energy saving in 802.11?

Idea: switch the transceiver off if not needed

4. Does power saving work in ad-hoc mode as well?

Yes.

Power Saving with Wake-up Patterns (Ad-hoc)



5. • What information does a TIM contain? What is a DTIM?

- Traffic Indication Map (TIM): List of unicast receivers
- Delivery Traffic Indication Map (DTIM): List of broadcast/multicast receivers

6. • How does a handover work in 802.11? Why is it so slow, compared to GSM or UMTS?

Scanning

– scan the environment, i.e.

☐ listen into the medium for beacon signals or

☐ send probes into the medium and wait for answers

=> time-consuming (scan all channels)

Physical Layer Variants and Extensions

1. • What are advantages and disadvantages of using higher frequencies?

Advantages - higher bandwidth

Disadvantage - cannot penetrate through because of lower wavelengths

2. • Why is the transmit power limited to 100 mW?

To limit the interference from adjacent transmitters.

Mobility Management

01 March 2021 14:41

Mobility Management in the Reference Models

1. • **What are the tasks of mobility management?**
 - Find (also called paging) and address a mobile away from home
 - Support “continuous” communication after change of location
 - Security: avoid misuse of mechanism
 - Privacy: hide location from others (FSSP)
2. • **Which protocol layer should handle the mobility problem? What are the implications of selecting one or another layer?**
 - Local mobility - moving inside the same subnet - MAC and PHY layer
 - Global Mobility - moving across subnets - Network, Transport and application layer.
3. • **What are the performance criteria for handovers?**
 - Delay to update the route
 - Packet loss due to handover
4. • **Why can the cooperation between layers improve the handover performance?**
 - Optimization of the performance of a certain layer mobility management approach using information from other layers or the integration of solutions from several layers.
 - Synchronization between layers is essential to provide a good handover performance.

Link Layer Mobility Management (Layer 2 and PHY)

- **Which mobility problems can be handled on the link layer? Which not!**

Mobility inside the same subnet (Local Mobility) can be handled at the link layer.
Global mobility cannot be handled.
- **What happens if a mobile moves from one WLAN coverage area to another? What is the main source for handover delay?**

A handoff process happens as below: (RSRR)

 - Recognition of the loss of the wireless connection
 - Search for and detection of a new adequate AP
 - Re-/Authentication with the new discovered AP
 - Re-/Association with the new discovered AP

Scanning

 - **How could this be accelerated?**

periodic scanning : the MN when detects weak signal, switches to a different frequency for sometime and scans for new AP while still not completely detached from old AP.
Information from neighbours : utilizing the information about neighbouring AP to reduce the number of channels that must be scanned.

Network Layer Mobility Management

- **How can mobility be supported by the network layer?**
 - Specific routes to end systems (per-host forwarding)
 - Multicast
 - Modification of the destination address
 - Tunneling
- **When does the sole update of routing tables make sense and when not? Why does it not scale?**

Change of routing tables is done to support global mobility of a MN. It does not make sense to use this method when there are many MN to be supported.
It does not scale because it required Distribution of routing information for every individual mobile (rather than subnets) everywhere
- **What is the problem with changing the IP address for mobiles due to mobility?**

Problems to find a mobile host without a constant address
Problems with application due to change of address
Security problem
- **What is the purpose of tunneling? What is the advantage of this over specific routes?**

Separation of a terminal/user identifier from a topological locator
It is scalable.

Mobile IP

1. • **What is the overhead imposed on data traffic with Mobile IP?**
 - Sub-optimal routing and increased end-to-end delay due to triangular Routing
 - Overhead due to encapsulation
1. • **What is the overhead for signaling with Mobile IP?**
 - Signaling load due to frequent BUs especially when MNs move at high speeds
2. • **When is triangular routing especially harmful? Can we solve this and how?**
 - o Higher latency and network load
 - o CN learns the current location of MN
 - o Direct tunneling to this location
3. • **What are the differences between the CoA-handling in the FA vs. handling in the MN itself?**
4. • **Which optimizations can be made in order to smooth data transport after a handover?**
5. • **What is the purpose of reverse tunneling? In which case is it helpful?**
6. • **What is the purpose of DHCP? What problem does it solve?**
 - Supplies systems with all necessary information, such as IP address, DNS server address, domain name, subnet mask, default router etc
 - Simplification of installation and maintenance of networked computers ; Enables automatic integration of systems into an Intranet or the Internet

Network Layer MM – Advanced solutions

1. • **Identify the limitations of Mobile IP. Which optimizations can you think of?**
 - Management of frequently moving stations.
 - Macro and Micro mobility.
2. • **What is the purpose of these optimizations?**
 - To handle local problem locally. Without involving the Home agent.
3. • **Which problem is addressed by Hierarchical MIP?**
 - Micro mobility support
4. • **What is the idea of HMIP?**
 - Movements inside the domain are controlled by the intermediate node, also called micro or intradomain mobility.
 - Movements between different domains are typically handled by MIP/home agent, also called macro or inter-domain mobility
5. • **Which problem is addressed by Cellular IP?**
 - Improves micro mobility strategy.
6. • **What is the idea of Cellular IP?**
7. • **Why do updates of the routing table make sense here?**
8. • **Can Cellular IP be generalized to non-hierarchical systems?**
 - Yes.
9. • **Which problem is addressed by Proxy MIP?**
 - Hide the identity of the network in which it is roaming in.
10. • **What is the idea of Proxy MIP?**
 - The idea is to provide the mobile node with an environment that always feel like home - done by the emulation of access gateway like the home network of the mobile node (give everything like the mobile node gives like home)

Criteria to compare mobility protocols (HPSRE)

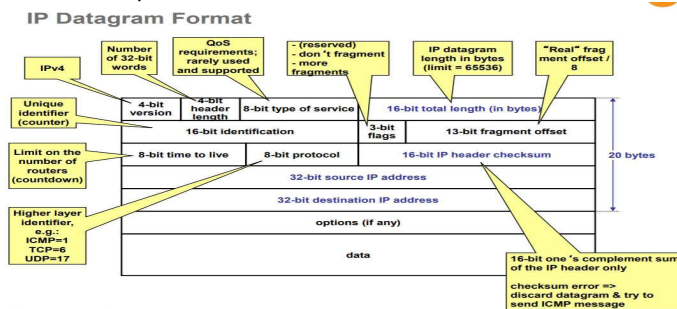
- Handover (HO) performance
 - Delay to update the route
 - Packet loss due to handover
- Protocol overhead: consumption of network resources
 - Scalability: protocol overhead in large networks with large numbers of mobile nodes
- Robustness: adaptability to different network conditions and failures
 - Ease of deployment: simplicity, suitability to different network scenarios

Transport protocol

06 March 2021 16:58

Internet Protocol Suite

- **What is the purpose of ARP?**
Address Resolution Protocol - resolves an IP address to Mac.
- **What is the difference of ARP from DNS?**
ARP (Address resolution protocol) which translates the IP address to its associated physical address.
Applications use the Domain Name Service (DNS) to convert hostnames (e.g. www.nokia.com) into IP addresses (88.221.214.58) and vice-versa
- **How does ARP differ from DHCP?**
ARP (Address resolution protocol) which translates the IP address to its associated physical address.
DHCP is used to automatically allocate an IP address from a defined range of numbers available in a network.
- **What is the purpose of encapsulation and decapsulation?**
To add necessary feature information to the actual data packet that will be used for secure, successful and correct transmission in the medium towards the destination application on the correct machine.
- **Why do we need multiple headers? Why not a single one?**
Each layer provides different services which are offered by various protocols. Each of these information is added to the data packet at different levels for a proper ordered processing at both the sender and receiver end.
- **What does the MTU mean? What is its purpose of the path MTU?**
Maximum transfer unit - Maximum IP packet size in bytes.
The smallest volume of data that can be transported over the complete path without being fragmented.
- **What is the purpose of the different header fields in IP? What is the purpose of the checksum and the protocol field?**



- **What is the purpose of ports used by TCP and UDP? How does it differ from IP addresses?**
Ports are used for application multiplexing.
IP address is used for identifying the server or the machine that the destination application is running on.
- **How are errors handled with UDP? Why is UDP not considered reliable?**
There is no error handling in UDP, the corrupted or out of order packets are simply discarded. UDP does not provide error correction and is therefore an unreliable protocol. In other words, delivery of packets is not guaranteed. UDP datagrams are transmitted without provision for an acknowledgment.
- **Why does the TCP/IP model provide different transport layer protocols?**
TCP/IP and UDP can be used appropriately based on the requirement of the application.
TCP/IP for connection oriented and reliable communication where packet loss is not tolerable.
UDP for applications where packet loss is tolerable.

TCP over wireless

- **What are the problems of the use of TCP over wireless links?**
 - TCP assumes congestion if packets are dropped - for wireless this could be transmission errors (fading) or mobility from one access point to another.
 - High latency of wireless systems prolongs the slow start period
 - The performance of an unchanged TCP degrades severely
- **Identify possible effects of high fluctuations of the channel capacity on TCP?**
TCP reduces data rate.
- **What is the difference in the impact of temporarily poor radio channels for the case that LLC is empty or uses some ARQ over the wireless link?**
- **Can you think of mechanisms to alleviate the problem?**
- **What is the advantage of splitting TCP connections in a wired and a wireless part, what's the disadvantage?**
 - transmission errors on the wireless link do not propagate into the fixed network
 - no changes to the TCP protocol for hosts connected to the wired
 - Internet, millions of computers use (variants of) this protocol**Disadvantage**
 - A handover between access points requires the migration of the TCP sockets and the TCP state (buffers, etc.) - latency
 - higher latency possible due to buffering of data within the foreign agent
 - loss of end-to-end semantics
- **Are there less intrusive solutions to the TCP-over-wireless problem?**
Yes, Snooping TCP.

TCP (Transport Control Protocol)

- **How does the MSS differ from the MTU?**
The MTU indicates the size of the complete TCP / IP packet. The MSS only specifies the space for the user data in the TCP / IP packet (MTU - (Ipheader + TCP header))
- **What is the purpose of the MSS?**
It limits the maximum size of a data that can be accommodated as a TCP segment.
- **What is the purpose of the different flags?**
URG: Urgent Pointer field significant - urgent data are outstanding
ACK: Acknowledgment field significant
PSH: Push Function - push to indicate prompt transmission of data
RST: Reset the connection
SYN: Synchronize sequence numbers
FIN: No more data from sender
- **What is the purpose of the checksum field?**
It is used to check if the received packet is corrupted or not. The TCP segment checks for the [TCP header + IP pseudo header + data] validity.
- **What is the purpose of the window size field?**
It is the advertised window size from the receiver. number of bytes, starting with the one specified in the ack. number field, that the receiver is willing to accept.
- **What is the purpose of DUPACKS?**
With DUPACKS, the receiver informs the sender about a missing segment when an out of order segment arrives from sender.
- **How does a TCP sender learn about overload in the network? Which messages and mechanisms are used for this?**
Dupacks - small problem
Timeouts - Severe problem
Congestion window size - This value will be reduced and sent back to sender by the receiver when there is overload at the receiver (Indirect learning at sender)
- **How does the TCP congestion control algorithm in the sender react to overload problems? What are the options?**
When there is an overload in the network, packet loss occurs.
 - When the packet loss is detected by timeout - Sender goes back to "Slow Start" Phase and sets the ssthresh to half the CWND
 - When the packet loss is detected by Dupacks - Sender employs "Fast retransmit" of the lost packet and halves the window size and sets the same to ssthreshold.
- **Why does TCP's response to timeouts differ from the response for DUPACKS? What is the purpose of this?**
Timeouts signify severe problem with the link and hence the sender reduces the data rate to give time for the link to recover
Dupacks signify a loss of single segment or out of order reception which is seen as a small problem and hence sender quickly re-transmits the missing segment for continued communication.
- **Can we find out the direction in which the congestion occurs?**
It is always towards the receiver.

Beyond TCP

- **What are the advantages of moving congestion control mechanisms into the user space?**
Flexibility and easy adaptability
- **What does HoL blocking mean?**
HoL arises due to in-order delivery requirement that creates delays when a packet is lost
- **What is the benefit of parallel multiple data streams?**
Reduced latency.
- **What is a bufferbloat?**
Is a problem that arises when there is huge buffers in front of overloaded links. Large buffers delay (TCPs) congestion mechanisms to learn about the overload and to slow down their data rate
- Without QoS mechanism in place this introduces increased delay and delay jitter especially harmful for latency-sensitive traffic as VoIP, gaming, etc
- **What are incast issues?**
Many-to-one simultaneous traffic flows create congestion and packet drops at a single node, e.g. switches, resulting in long delays
- **What are the benefits of SPDY?**
Reduces required bandwidth through header compression
Targets reduced latency - fast access to web pages
Multiplexes several data streams (requests) over several TCP Connections - solves HoL problems of TCP
header compression - reduces bandwidth consumption.
- **What are the benefits of QUIC?**
Easier adaption. Handles congestion and flow control in application layer / user space

6. • Are there less intrusive solutions to the TCP-over-wireless problem?

Yes, Snooping TCP.

7. • What benefit can TCP draw from lower layer information? How can it be used?

Mobile hosts can be disconnected for a longer time

scheme is independent of data

It can be used for Transmission/timeout freezing of TCP in case of link disconnection.

problems of TCP

header compression - reduces bandwidth consumption.

6. • What are the benefits of QUIC?

Easier adaption. Handles congestion and flow control in application layer / user space

Reduced Latency

7. • What are the benefits of MPTCP?

Macro Diversity

Enhances system capacity, latency, reliability and reduces load on congested paths

TCP in wireless	Problems	Fading - data rate decreases.
	Solution	I-TCP
		Snooping TCP
		Mobile TCP
		Timeout freezing
		Forced re-transmit
		selective re-transmit
TCP	packet loss detection	Timeout
		Dupacks
	Flow control	Slow start
		Congestion avoidance

Quality Of Service

08 March 2021 16:16

QoS Basics

- **What are the QoS attributes of a connection?**
Latency, Jitter, throughput, packet loss
- **What are the basic functions for the provisioning of QoS?**
Admission control, traffic classification, traffic conditioning, scheduling, overload control
- **Name the strategies for QoS provisioning**
Reservation, differentiation, over provisioning
- **Which applications have the toughest QoS requirements?**
Applications which are delay intolerant and error in tolerant. Eg - interactive games.
- **Name measures to minimize the error rate over wireless links**
Employ high SINR, robust modulation schemes and high FEC redundancy.
- **Name the measures to minimize the average delay over wireless links**
By trying to do it right at the first time. Ensure high SINR, robust modulation schemes and some FEC redundancy to minimize frame errors.
- **Name the measures to maximize the throughput over wireless links**
Considering tradeoff between TxPower, modulation scheme and redundant coding: higher order modulation scheme increases throughput and reduces delay for burst transmissions but increases risk of errors
- **How can you compute the maximal throughput of an end-to-end connection from its constituents?**
- **How compute the end-to-end delay for a connection for the case of zero transmission errors?**
Delays caused by other factors such as processing delays, Scheduling, Packet loss and retransmission, System load and deterioration of link quality.
- **What are the reasons of delay jitter, what the reasons for packet loss for wired and wireless communication?**

Wired	Wireless
Latency - Transmission delays, processing delays Jitter - shared resources. Packet loss - finite buffering capacity (congestion points).	Latency - processing for channel coding Jitter - scheduling, packet loss, System load, Channel quality Packet loss - maladjusted modulation, coding, TxPower, bad signal, low SINR

QoS Functions and Common Concepts

- **Which functions in a QoS-enabled network ensure that an application does not exceed its throughput requirements?**
Traffic shaping, Policing, Metering, Marking and re-ordering.
- **How does the token bucket scheme helps in the limitation of the user throughput? How does the System support averaging?**
Token bucket scheme employs Metering which allows traffic only at the rate at which the tokens are made available. Each incoming packet needs a token and thus, the incoming traffic is limited to the token rate that can be set to the desired throughput.
Averaging is supported by Random early detection.
- **What is the purpose of explicit packet dropping in routers?**
- **How packet dropping influences the performance of TCP connections?**
- **What are the ingredients of the TOS field? Why are multiple fields needed?**
TOS has three fields -
Precedence field - tells the priority of the packet.
TOS: denotes how a device should trade off between throughput, delay, reliability and cost for the packet.
MBZ: Must be zero.
- **Why is the TOS field mostly relevant to the network as such rather than the end system?**
TOS is a IP packet field and it does not depend on source or destination addresses.
- **Why do networks differentiate between edge routers and core router? What's the purpose and what is the difference in behavior?**
Edge and core routers are differentiated to give focus on certain functionalities to each type of the router.
Edge routers - apart from routing also handles security with strict admission controls. Also police and marks packets admitting to the network.
Core routers - concentrate mainly on forwarding huge amounts of data towards the destinations as necessary to cope with transient congestion within the network.

QoS over Wireless

- **Why is QoS considered a cross-layer issue?**
QoS is an important factor in every layer and its implementation is different in each layer.
- **What influences the QoS of a wireless link? Can the physical layer adapt or influence the QoS? How? Are there limits of this?**
Fading, mobility, etc. caused by
- Highly varying signal quality and path loss (due to none-line-of-sight, multipath, shadowing, etc.)
- Highly varying interference from others
Yes. By adapting modulation and coding schemes, use multiple antennas simultaneously, increase transmit power.
Yes. Higher Modulation - Bit error increases
Higher power - interference to other users.

QoS Strategies: IntServ, DiffServ, Overprovisioning

- **What are the three main strategies for QoS provisioning in a network?**
Intserv, diffserv, over provisioning.
- **Why does IntServ not scale? What is the problem of packet handling per individual flow? Hint: think about the implementation of rate control for individual flows.**
For each flow, in each IP packet - has to be handled by each router in the data path individually- so basically, routers should take a look at the IP address and TTL and quickly check up routing basics of a packet but now it should check QoS parameters as well like checking if the packet is within the specification - has the packet data flow has reached the capacity- is it running at the limit or below - lot of effort considering numerous number of data flows through a router.
- **Why does RSVP use soft-states? What does it mean?**
To release the resources that were reserved for a flow. If not released, then the resources are reserved for a lifetime even if they are not used (Eg- mobile terminal dies).
Soft state means the path reservation for the desired QoS has to be refreshed from time to time - this is done by some reservation refresh messages and path messages.
- **How does RSVP handle the case that a router is not supporting it?**
If the path message arrives at a router that does not understand RSVP, that router forwards the message without interpreting the contents of the message and will not reserve resources for the flow.
- **What are the common entities in routers that are present in both IntServ and DiffServ systems?**
Metering, Traffic shaping, Marking.
- **What are the major differences between IntServ and DiffServ wrt**
- **Provided QoS requirements**
- **Separation of flows**
- **Effort for edge and core routers**
- **How can we ensure that QoS works over multiple networks and multiple providers?**
To permit services which span across domains (multiple networks)
- Establish Service Level Agreements (SLA)
including Traffic Conditioning Agreements - TCA
- Common service provisioning policy
- Resource Management is performed at two levels
- Inside administrative domains
- Between neighboring domains (Bandwidth Broker - BB)
Concatenation of bilateral agreements leads to end-to-end QoS delivery paths
- **What is the difference between metering and marking?**
Metering detects the packets that are in-profile and out-profile
Marking, marks the packets that are out of profile for dropping in case of congestion or it is scheduled with low priority.
- **Under what circumstances can overprovisioning be applied?**
Appropriate where resources are abundant (typically not true for air interface) or max. traffic volume is known (may hold for access network).
When resources are limited, limit the amount of input traffic.

QoS Signaling

- **What's the idea behind NSIS?**
To have the virtual signal network on the lower layer and put on top the signalling application, so all the signalling application can use the same signalling network.
- **For which signaling purposes can NSIS be used?**
Receiver- and sender-oriented reservation
- **Does NSIS support secured signaling?**
Yes, with extra layer called - Transport security layer.
- **Does it use UDP or TCP for signaling?**
Both.
- **What is the purpose of the GIST sublayer?**
GIST (General Internet Signaling Transport protocol)
- Common signaling transport service for different signaling applications

QoS in 802.11

- **DCF extensions for QoS provisioning:**
- **What is the idea of the DCF extension? Which QoS strategy does it implement?**
Differentiation of access to different traffic classes.
Diffserv.
- **How does the extended version of DCF work?**
We use different DIFS Values called AIFS - increase the value (waiting timer).
Smallest DIFS for higher priority traffic. In addition, we play around with the CW size for priorities whenever the back off is used.
- **What are the parameters to differentiate traffic in EDCA?**
 - o AFIS Arbitration Inter Frame Space
 - o CW Contention Window (min & max values)
 - o TXOP Transmit opportunity (limit on access time to channel)

simultaneously, increase transmit power.
 Yes. Higher Modulation - Bit error increases
 Higher power - interference to other users.

1. • **Which PHY strategies are appropriate for the maximization of the throughput?**
 Select the MCS (Modulation and Coding Scheme) that maximizes the throughput at minimal cost to the radio system as such.
2. • **Which PHY strategies are appropriate for the minimization of the error rate? What are the drawbacks of the different approaches?**
 Select highest MCS that meets the given BLER target (here 10⁻⁵) for the given SINR (link condition) at minimal cost.
3. • **What is the benefit of ...**
4. • **What would be a good strategy to optimize the system capacity of the cellular system as a whole? On which layer would this apply?**
 There are lots of choices over multiple layers to choose from:
 The best combination of selected solution depends on the specific user requirements as well as the specific radio situation!

QoS and Mobility

1. • **What is the conflict between mobility management and QoS?**
 - High variance in quality and capacity of wireless channel due to movement of user/equipment (fading, shadowing, etc.)
 - After movements, the user has to reserve resources again - latency
2. • **What are possible solutions to this conflict?**
 - Coupling between QoS and mobility mechanisms
 - fast reservation and release of resources
3. • **What are the pros and cons of different approaches to the integration of QoS and mobility management**

- **What are the parameters to differentiate traffic in EDCA?**
 - AFIS Arbitration Inter Frame Space
 - CW Contention Window (min & max values)
 - TXOP Transmit opportunity (limit on access time to channel)

- **Provide an example for the parameters for 4 traffic classes**
 - Best effort (AC_BE)
 - Background (AC_BK)
 - Video (AC_VI)
 - Voice (AC_VO)

– **With which parameter settings can an EDCA-enabled device outperform a non-QoS enabled device?**

- **What are the limits of the DCF extension with respect to QoS?**
 - The relative performance from this implementation is not guaranteed or easy to control.
 - The performance is NOT proportional to the backoff factor ratios
 - It depends on the number of contending stations

- **What is the purpose of TXOP?**
 - It is the maximum time the station may use to transmit its data.
 - Purpose - To resolve virtual collisions by the scheduler before transmission attempt.

2. • **HCCA**
 - **Which QoS strategy does it implement?**
 - intServ.
 - **Which are the important functions for its implementation?**
 - Provides policing and deterministic channel access by controlling the channel through the HC (Hybrid Coordinator)
 - Admission (or rejection) of stations based on Traffic Specification (TSPEC)
 - HC derives schedule to provide the guaranteed QoS requirements
3. • **How do the extensions react to changing channel conditions?**
 - No support

Self-Organizing networks

03 March 2021 20:45

Definitions and Properties

- **What's the idea of self-organized (SO) systems?**
Autonomously acting individual systems performing local programs and acting on local data but participating on a global task showing an emergent behaviour
- **Why is SO so popular for modern systems?**
 - Autonomous optimization and repair
 - Automation of control
 - Scalability
 - Absence of single point of failure
- **What are the properties of SO systems?**
No central control
Emerging structures
High resulting complexity
High scalability
- **What are the limits of SO systems?**
Loss of determinism
High effort for evaluation and Testing
Resulting in high effort for Development
- **What does emergence mean?**
Emergent behaviour of a system is provided by the apparently meaningful collaboration of components (individuals) in order to show capabilities of the overall system (far) beyond the capabilities of the individual components.

Design of Self-organized Systems

- **Identify important design principles of self-organized systems**
Positive and negative feedback
Interactions among neighbors and with the environment
Probabilistic techniques
- **Identify the design paradigms for algorithms of self-organized Systems**
Paradigm #1:
Design local behaviour rules that achieve global properties
Paradigm #2:
Do not aim for perfect coordination: exploit implicit coordination
Paradigm #3:
Minimize long-lived state information
Paradigm #4:
Design protocols that adapt to changes
- **How are these paradigms applied to communication protocols?**

Challenges and Limits

- **What defines the Edge of Criticality? Why is it such an important issue?**
The edge between
 - order (i.e. coordination, patterns) and
 - chaos (stochastics and variance)It is important because - Life, networks, the brain, the universe operate best at the "edge of chaos". But at the same time are extremely sensible to unforeseen changes.
- **What are the advantages and disadvantages of operating close to this critical point? What are the properties of systems operating at that point?**
Advantage: Life, networks, the brain, the universe operate best at that point
Disadvantage : are extremely sensible to unforeseen changes
- **Identify the points of criticality for different applications as TCPs congestion control, RED, 802.11 DCF and the collision avoidance of UAVs**
- **Identify ways to reduce the risk of criticality for different SO algorithms**
- **Discuss how the term "overengineering" influences the risk of criticality.**

Limitations of SON

- Controllability
- Predictability vs. scalability
 - Software development
 - New software engineering approaches are needed
 - design and tuning of self-organized algorithms is highly complex
 - System test
 - Handling of stochastic behavior
 - Cross-mechanism interference
 - composition of multiple selforganizing mechanisms can lead to unforeseen effects
 - stability is an issue

Applications of SO with examples and important aspects

Medium Access (CSMA / CD 802.11 DFC MAC)

Why is it an SO ?

- Its a SO because it has a feedback and based on feedback it employs back off accordingly.
- Its a SO because communication is solely with the environment to gain information.

Which aspect makes it SO?

- The aspect of back off timer adjusting itself to the environments response or feedback - if free, sends data ; if busy, waits random time ; if collision, exponentially increasing contention window max size - is considered Self organizing.

Which are positive and negative feedbacks?

- Neg - collision
- Pos - medium free

Where is communication in it ?

Implicit communication via back off mechanism.

What aspect is stochastic in the example ?

- Stochastic because of the randomness of the appearance of diff packets in diff mac layers.

Which level of level 4 paradigm is implemented in the example ?

Level 2- keep the communication implicit.

TCP congestion control

Why is it an SO ?

- Reaction from sender is diff based on the feedbacks. 3 dupacks - half the congestion window; timeout - Dramatically reduce data rate (these are local actions resulting in global solution)

Which aspect makes it SO?

- The aspect of changing the CWND size as per the feedback and contributing to global successful traffic maintenance.

Which are positive and negative feedbacks?

- ACK(Pos feedback), DUPACK(Neg Feedback), no ack(Neg feedback)

Where is communication in it ?

- The ack and dupack packets

What aspect is stochastic in the example ?

- Random early detection - negative feedback would be some packets missing and positive feedback is everything working fine.

Which level of level 4 paradigm is implemented in the example ?

Level 1- Cope with changes.

LEACH: Low-Energy Adaptive Clustering Hierarchy(Cluster Head Election in Sensor Networks)

Why is it an SO ?

LEACH is a self-organizing, adaptive clustering protocol that uses randomization to distribute the energy load evenly among sensors in the network

Which aspect makes it SO?

The nodes decide themselves for sensor data collection or any other sensor activities based on the energy levels of each node which brings about global energy saving and even distribution among all nodes.

Which are positive and negative feedbacks?

- To agree on a cluster head proposal by other nodes - positive
- Not to agree - negative

Where is communication in it ?

The broadcasting of cluster head information to the network

What aspect is stochastic in the example ?

Randomization to distribute the energy load evenly

Which level of level 4 paradigm is implemented in the example ?

The level 3 - adaptive protocol.

Inter-system Load Balancing

Why is it an SO ?

Which aspect makes it SO?

UE is pulled to different cells because of the forces exerted by the cells on the UE depending on the appropriateness of the cells

Which are positive and negative feedbacks?

- Criteria/forces:
 - supported QoS in target cell (user view)
 - free capacity of target cell
 - cost per bit in target cell
 - migration penalty and handover cost

Where is communication in it ?

What aspect is stochastic in the example ?

Which level of level 4 paradigm is implemented in the example ?

Collision Avoidance for UAVs

Why is it an SO ?

Emergence of roundabout

Which aspect makes it SO?

Feedback based mechanism from the environment where an UAV sends a broadcast signal. Receiving UAVs apply a force/torque model to adapt own movement

Which are positive and negative feedbacks?

No broadcast information is positive which says no UAV in its path

Existing of a broadcast information from an UAV in its path is negative which results in a detour to avoid collision.

Where is communication in it ?

Each UAV periodically broadcasts own coordinates and movement vector

Each UAV periodically broadcasts own coordinates and movement vector

What aspect is stochastic in the example ?

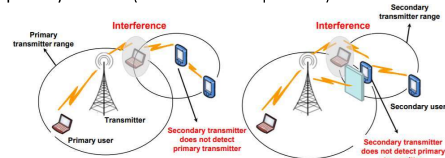
The physics implemented here always has some stochastic aspect.

Which level of level 4 paradigm is implemented in the example ?

Introduction

- What is the idea of cognitive radio? Why is it important?**
 A cognitive radio is a radio that can change its transmitter parameters dynamically based on interaction with the environment in which it operates
 Allocation of temporarily unused spectrum band.
 Utilization of the allocated band for communication as long as no primary user appears on his band and no (or limited) interference is produced for primary users in the neighborhood.
 - Limited spectrum
 - Inefficiency in spectrum usage
 Cognitive radio technology is required to enable better utilization of the unused licensed spectrum.
- What defines a primary network, what a secondary network?**
 Primary network - Networks with access right to certain spectrum bands. Users have the right to operate in licensed spectrum.
 Secondary network - Do not have license to operate in the spectrum band.
- What defines a spectrum hole?**
 The unused portion of a spectrum is called a spectrum hole.
- What are the main tasks and challenges of cognitive radios?**
 - Determination of a spectrum hole (Spectrum sensing)
 - Selection of the best available channels (Spectrum decision)
 - Co-ordination of access with other users (Spectrum sharing)
 - Detection of the appearance of licensed user - vacate the channel (Spectrum mobility)
- Why do cognitive radios need to be configurable?**
 For transmission and receipt of data on a variety of frequencies and using various radio access technologies
- What are the differences between cognitive radio networks and traditional wireless networks?**

Spectrum Sensing

- Why is accurate spectrum sensing for cognitive radios important?**
 To avoid the use of spectrum where the primary network is impacted by interference
- Explain the hidden terminal problem for cognitive radio networks**
 Secondary user can only sense the primary transmitters, while interference occurs at the primary receiver (hidden-terminal problem)
 
- What are the advantages of cooperative sensing? How does cooperative sensing solve the hidden terminal problem?**
 - Incorporating information from multiple secondary users to detect primary users
 - More accurate detection than that based on single secondary user observations
 - There are many secondary users sharing information among them so if one sec user misses to locate a primary transmitter in its range, the other sec user can locate it in its range and thus solving the hidden terminal problem in the network.
- What is the idea of matched filter sensing? What are the prerequisites?**
 - Check of correlation of received signal with a given template, e.g. a pilot, that allows detecting the presence of a known signal (Maximize SNR)
 - Optimal detector when all characteristics of the primary user signal are known
- What does energy detection mean? What is its advantage?**
 - It is the technique of detecting the primary network's transmitter based on its signal power (or energy of the signal). It is used when there is no information about the primary signal is known.
 - Low implementation and computation and complexities.
- How does cyclostationary feature detection work?**
 It is a technique to detect the primary user signal based on the fact that mean and autocorrelation of modulated signals exhibit periodicity (because modulated signals are coupled normally with sine wave carriers, pulse trains, repeating spreading)
- What is the idea of interference temperature management?**

IV. INTERFERENCE BASED DETECTION

Transmitter detection and cooperative detection methods focus on reducing the interference to the primary transmitter. Interference based detection shifts its focus to minimize the interference to the primary receiver irrespective of the primary transmitter's operation. The signal power received at

Spectrum Decision, Sharing and Mobility

- What are the advantages of a centralized secondary network architecture?**
 Global knowledge and centralized control of the spectrum allocation and access.
- What are the differences between overlay and underlay?**
 - Overlay spectrum sharing
 Secondary users access unused spectrum holes
 Avoid interference to primary users
 - Underlay spectrum sharing
 Secondary users stay within acceptable interference temperature
 Often based on CDMA and spread spectrum techniques
- Which criteria are relevant for the effective selection of the spectrum hole?**
 - Bandwidth of spectrum hole
 - Path loss (interference generated) towards primary users
 - Path loss between secondary users
 - Characteristics of primary transmission (Radio technology, coding, detectability)
 - Expected holding time of channel
- What is spectrum mobility?**
 The process when a secondary user changes its frequency of operation, also called spectrum handoff
- When can a secondary transmitter use a frequency band at the same time as a primary transmitter? What are the requirements for this?**
 When the secondary user transmission interference with primary user transmission is limited (defined by the interference temperature).
- For which kind of cognitive radio approaches is the application of spectrum preading and CDMA especially useful? What is the advantage?**
 Underlay spectrum sharing

Cognitive Radio MAC Protocols

- What are the benefits of having multiple radios for cognitive stations?**
 It is much simpler to allow for parallel/ Independent implementation of CCC
- What are the advantages of centralized MAC schemes?**
 Central entity manages, synchronizes and coordinates operations among secondary users
- What is the functionality of the common control channel? What are the challenges?**

Tasks:

 - Exchange of sensing information
 - Coordination (with a central entity) to organize the spectrum allocation
 - Transmitter-receiver handshake

Problems:

 - Fixed CCC is infeasible (CCC must be vacated when a primary user appears on it)
 - CCC may vary over time and between users due to spatial distribution
 - If no CCC is allocated, transmitter-receiver handshake becomes a challenge
- What are the ideas of CSMA-MAC and SYN-MAC? How is the common control channel implemented in the two schemes?**

CSMA-MAC:

 - csma technique for channel access by both PU and SU.
 - Operates in underlay and overlay
 - Extended sensing period

SYN-MAC:

 - Each channel has a time slot for control messages (control slot)
 - Each node has two radios, one for listening to control messages and one for transmission of data

CCC:

 - In CSMA-MAC, a single transiever, CCC and data on same frequency.
 - In SYN-MAC, CCC is multiplexed in time, between different channels.
- Identify the differences between CSMA-MAC and SYN-MAC and identify under which situations the individual schemes show their strengths**
 High and low load

Software Defined Radio (SDR)

- What is the idea of SDRs?**

The idea is to capture the signal from the antenna and directly feed it into an ADC (Analog-Digital) converter and do all the remaining processing in a digital domain.

Transmitter detection and cooperative detection methods focus on reducing the interference to the primary transmitter. Interference based detection shifts its focus to minimize the interference to the primary receiver irrespective of the primary transmitter's operation. The signal power received at the primary receiver reduces exponentially with the distance, until it reaches a level of noise floor. At this point, although the primary transmitter is operating, the primary receiver treats this communication as simply noise and not transmission; hence the secondary user can utilize the channel since no interference is introduced to the primary user's communication, as the primary receiver is not in the receiving mode. An interference cap is introduced above the maximum noise level, known as interference temperature; below this level the primary receiver will treat this transmission as simply noise. If the detected primary signal level is below the interference temperature, the secondary user may utilize that channel. Further if the transmission power of a CR remains below the interference cap, it may utilize any frequency parameters of its choice. A main challenge faced in implementation of this method is the receiver interference temperature determination.

The idea is to capture the signal from the antenna and directly feed it into an ADC(Analog-Digital) converter and do all the remaining processing in a digital domain.

2. • **What are the advantages of SDRs**
 - Reconfigurability by means of reprogrammable software
 - Implement only most necessary things in HW
 - Instead of HW components use DSPs, FPGAs, or GPPs
3. • **What are the disadvantages and limits of SDRs?**
 - Fast and accurate spectrum sensing equipment and mechanisms required.
 - Like nanoseconds delay requirement - interface delay

4. • **How do SDRs compare to traditional radio implementations?**

<u>Traditional radio</u>	<u>Software defined radio</u>
Tailored to fixed requirements and operating conditions	Reconfigurability by means of reprogrammable software
Highly optimized off-the-shelf and application-specific components	Implement only most necessary things in HW
Static design: modifications impossible after fabricated	Flexible - can be modified via software.

Ad hoc networks

12 March 2021 16:59

Basics

1. • **What are the various forms of ad-hoc networks?**
Wireless mesh networks, Vanets, Wireless sensor networks, Delay Tolerant Networks,
2. • **How are ad-hoc networks formed? What are their constituent components?**
Ad hoc networks are formed spontaneously without any infrastructure. The network constitutes mobile devices that communicate through packet forwarding. Ad hoc networks are self-organizing as there is no central managing entity.
3. • **What characteristics are exhibited by ad-hoc networks?**
 - Spontaneous federation of wireless devices
 - No infrastructure (base station / access point), no backbone
 - Devices can be mobile
 - Packet-based forwarding
 - Mobile device also serve as routers
 - Routes between devices can span multiple hops
 - Ad hoc networks are self-organizing
 - No central components
4. • **What advantages do ad-hoc networks offer?**
In ad hoc network, node acts as a router to send and receive the data. An advantage of the system is robustness, flexibility and mobility
5. • **Which applications can benefit from the use of ad hoc networks?**
Military, DTN, Disaster communication, Extension of cellular network beyond coverage, sensor applications.
6. • **Discuss the advantages and disadvantages of different routing strategies for ad hoc networks**
 - Topological**
 - **Pro-active**
 - **Reactive**
 - **Hybrid**
 - Graphical**
 - **Greedy** - Disadvantage - shortest path on which destination is not reachable
 - **Face routing** - Advantage - solves the problem of loops forming faces.
 - **Hybrid**
 - Delay Tolerant Networks** - Disadvantage - Works only for networks which accept maximum delays.
7. • **What is the idea of delay-tolerant networking?**
Idea: Nodes are not continuously connected to each other, rather use communication opportunities due to the mobility of nodes (message ferries).
8. • **What are the advantages and disadvantages of geographical routing**
 - Advantage - No need for topology information
 - Disadv - destination location must be known to the source

Reactive Routing – AODV

- Which routing table entries are established with the RREQ and RREP messages, respectively?
RREQ - source node, source SQN, next hop, hop count, TTL
RREP - destination node, destination SQN, next hop, hop count, TTL
- How is the forwarding and how the reverse path established in the routing tables?
- What is the purpose of the sequence number and the hop count maintained in the routing tables?
SQN - denotes the freshness of its routing information in the network's routing tables.
Hop count - incremented by each intermediate node before forwarding RREQ - it denotes the number of hops to reach the node(destination/source) specified.
- What is the purpose sequence number in RREQs and RREPs?
A node increases its own SQN whenever a new route needs to be established (route discovery) or updated (route maintenance), e.g. due to link changes or outdated routing information
- How are multiple RREQs handled by the receiving node? What is the meaning of sequence number and hop count for this handling?
- How are link failures detected? What are the mechanisms to repair the link or route?
 - Absence of link layer ACKs for forwarded data
 - Absence of hello messages
- What is the purpose of the RERR message? Which routing tables are modified with this message?
 - It is sent to notify the source about a link failure.
 - Invalidate all destinations in routing table (hop count – infinity) that are part of the RERR (unreachable destinations)

MANET Routing

1. • **What are the challenges of routing in multi-hop ad-hoc networks?**
 - Highly dynamic topology
 - Mobility of devices
 - Varying quality of wireless channel (fading)
 - Partitioning and merging of ad hoc networks possible
 - Limited battery capacity of mobile devices
 - Additional battery drain due to (e.g.) routing functionality
 - Limited bandwidth
 - Additional bandwidth required for routing and MAC functionality
 - Wireless medium is broadcast medium
 - Hidden and exposed terminals
 - Asymmetric/unidirectional links
 - Different quality in both directions
 - Time synchronization difficult
 - Problem for low power modes (e.g. sleeping periodically)
 - Security mechanism hard to apply
 - Every device must be able to forward packets → no encryption of routing headers
2. • **What are the differences between link state and distance vector routing protocols?**
 - distance vector - Disadv - huge routing table information
 - Link state routing(OLSR) - Adv - small routing table information
3. • **What are different classes of ad hoc routing protocols?**
Topological routing and graphical routing.
4. • **What are differences between reactive and proactive routing protocols?**
 - Proactive**
 - Routes are calculated periodically and before needed
 - Keep routing information in all nodes up-to-date
 - Reactive**
 - Routes are only calculated when needed
 - Do not keep routing information in all nodes up-to-date
5. • **What are the motivations to have hop count and seq. number in RREQ of AODV?**
 - Hop count ensure that duplicate routes are eliminated (limits flooding)
 - SQN prevents confusion between old and new RREQs (freshness); Loop prevention during route discovery
6. • **Are RREQ and RREP broadcast messages or unicast? Why?**
RREQ is broadcast - source does not know where the destination is so discovery process in all directions until it finds the destination.
RREP is unicast - trace back to the source is marked in the route discovery process by RREQ so follows that route.
7. • **How does a node behave when it receives more than one RREQ?**
Check the SQN of the RREQ, if same then retain the route with lower hop count
If SQN is updated(new), then update the new route irrespective of the hop count value.
8. • **What is the idea of greedy geographical routing?**
Forwards data to the node that is located geographically near to the source and which potentially lead to the destination.
9. • **What types of information are needed to apply geographical routing protocols in ad hoc networks?**
Destination location must be known to the source

Cellular networks

13 March 2021 20:00

GSM, GPRS & EDGE

1. • **What is the idea behind GSM? What is its access scheme? Why is FDD preferred over TDD? Describe the mixture of FDMA and TDMA used for GSM.**

The concept of GSM emerged from a cell-based mobile radio system. FDD has inherently low latency compared to TDD. In case of FDD, both UL and DL spectrum are transmitting and receiving simultaneously.

2. • **GSM handover: what's the idea? What's the difference to a handover in a WLAN system?**

Change the cell while communicating

GSM - make before break so HO is faster.

WLAN - make after break.

3. • **Is there a handover if a GSM mobile does not have an active call? Why or why not?**

No. Unnecessary signalling. When there is no active call, paging happens for location update of the mobile user.

4. • **How does authentication work in GSM? Who or what is authenticated? How is the integrity of the shared key ensured?**

Challenge-Response:

- Authentication center provides RAND to Mobile
- AuC generates SRES using Ki of subscriber and RAND via A3
- Mobile (SIM) generates SRES using Ki and RAND
- Mobile transmits SRES to network (MSC)
- network (MSC) compares received SRES with one generated by AuC

5. • **What is the difference between hard and soft handovers?**

Soft handover - a mobile will be served by two Base stations before breaking up with one of them.

Ensuring no loss of service at any point.

Hard handover - a mobile will be detached from one serving station and re attached with another one.

Loss of service possible.

6. • **What is the difference between make-after-break and make-before-break?**

7. • **What are the main differences between a GSM and a WLAN system? Compare the two systems with respect to media access, radio resource management and handover execution.**

Spectrum management and utilization

- coverage and interference management due to cell planning
- capacity and load management due to admission control and QoS support
- flexible cell size simplifies cost-efficient nationwide coverage

Mobility management

- fast, lossless HO due to make-before-break

General control structures and control philosophy

- high reliability and QoS guarantees due to centralized/infrastructure-based management and control of all resources

Energy

- high energy cost on network side, low cost on mobile due to passive cell camping instead of active association, paging mode and sleep cycles

Customer relations

- monthly/bi-yearly contracts, pay per service
- security due to preshared credentials

Implementation

- simple implementation of TDMA, e.g. with GNUradio and SDR

8. • **Spatial reuse in cellular systems: compare a reuse of 3 with a reuse of 9 with respect to capacity (data rate) and quality (error rate) for GSM voice and GPRS/EDGE.**

3.

Spectrum divided to 3 parts

Smaller distance- higher interference.

Capacity per cell is bigger - quality is smaller.

8.

Spectrum divided to 7 parts

Frequency space smaller

Adv - low interference but low capacity as well, per cell.

LTE (Long-Term Evolution)

1. • **What is the advantage of a common IP-based packet core over separated networks for packet-switched and circuit-switched traffic?**

2. • **What is the purpose of opportunistic scheduling? What is the disadvantage?**
Prefer mobiles with good channel quality => maximizes system throughput
Ensure fairness for users with continuously poor channel quality

3. • **How does interference co-ordination work? What is the idea?**

To improve the interference in the cell edges, adaptive reuse of cell frequencies are done. By this the cell edge users and cell center users will have different frequencies for use.

4. • **What is the goal of self-organization in LTE?**

minimize OPEX by automation of planning, optimization and repair of the network

5. • **What are the use cases for self-organized network management in LTE?**

Use Cases for Self-organized Network Management

- Physical cell-ID automatic configuration (PCI)
- Automatic Neighbor Relation (ANR)
- Coverage and capacity optimization (CCO)
- Inter-cell interference coordination (ICIC)
- Random Access Channel (RACH) optimization
- Mobility load balancing optimization (MLB)
- Mobility robust optimization (MRO)
- Energy saving (power on/off)